

# Clinical Risk Assessment Report: Patient 78

## 1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 55.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.51x the diagnostic threshold

### Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET), which elevated the patient's absolute risk by +8.5%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

## 2. Key Pathological & Imaging Drivers (Elevating Risk)

**Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET)**      **Value: 2.59 [Top 3%] (+8.5%)**  
*Points to continuous, elongated bands of high-density or highly active tumor tissue.*

**75th Percentile Tumor Density (CONVENTIONAL\_HUQ3)**      **Value: 0.83 [Top 20%] (+4.9%)**  
*Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.*

**Large Zone Emphasis (CT) (GLZLM\_LGZE.CT)**      **Value: -0.39 [Bottom 4%] (+4.6%)**  
*Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.*

**Maximum Tumor Density (CONVENTIONAL\_HUmax)**      **Value: 3.87 [Top <1%] (+4.2%)**  
*Reflects the most dense solid components or calcifications within the primary tumor lesion.*

**Tumor Pixel Correlation (CT) (GLCM\_Correlation.CT)**      **Value: 1.23 [Top 8%] (+3.4%)**  
*Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.*

**Short Run Low Gray-Level Emphasis (CT) (GLRLM\_SRLGE.CT)**      **Value: -0.28 [Bottom <1%] (+3.3%)**  
*Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.*

**Large Zone Low Gray-Level Emphasis (CT) (GLZLM\_LZLGE.CT)**      **Value: 2.38 [Top 2%] (+3.0%)**  
*Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.*

**Carcinoembryonic Antigen (CEA)**      **Value: 2.6 [Avg (39th %ile)] (+2.5%)**  
*Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.*

## 3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

